1. Distributed Consensus & State Machine Replication

The Raft consensus algorithm decomposes replicated state machine coordination into three subproblems: leader election, log replication, and safety. A cluster elects a single leader per term via randomized election timeouts (150–300 ms). Once a log entry is replicated to a majority quorum of nodes, it is committed and durable across minority node failures.

Figure 1.1: Raft state transitions between Follower, Candidate, and Leader across election terms.

2. Hybrid Retrieval & Reciprocal Rank Fusion

Modern retrieval-augmented generation (RAG) systems combine dense vector similarity with lexical BM25 keyword scoring using Reciprocal Rank Fusion (RRF). Multimodal pipelines embed both prose chunks and extracted diagrams into a shared 768-dimensional vector space.

Figure 2.1: Hybrid dense + BM25 retrieval pipeline with visual figure extraction.